Numerical Linear Algebra II: 07 Principal Component Analysis

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Ostrava

Data Matrix and Basic Sample Statistics

Dataset with m **samples** (rows) and n **features** (columns), stored in $\widetilde{\boldsymbol{X}} \in \mathbb{R}^{m \times n}$:

$$\widetilde{\boldsymbol{X}} = \begin{pmatrix} - & \tilde{\boldsymbol{x}}_1^T & - \\ & \vdots & \\ - & \tilde{\boldsymbol{x}}_m^T & - \end{pmatrix} \in \mathbb{R}^{m \times n}.$$

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For two features $\boldsymbol{a} = (a_1, \ldots, a_m)^T$ and $\boldsymbol{b} = (b_1, \ldots, b_m)^T$, each sample is $\tilde{\boldsymbol{x}}_i^T = (a_i, b_i)$:

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- **Sample Covariance:** $\mathrm{Cov}(A, B) = \frac{1}{m-1} \sum_{i=1}^{m} (a_i - \mu_A)(b_i - \mu_B)$

Centering the Data Matrix

Let $\boldsymbol{\mu} \in \mathbb{R}^{1\times n}$ be the row vector of column (feature) means:

$$\boldsymbol{\mu} = \left(\frac{1}{m}\sum_{i=1}^{m} \tilde{x}_{i,1},\ \frac{1}{m}\sum_{i=1}^{m} \tilde{x}_{i,2},\ \ldots,\ \frac{1}{m}\sum_{i=1}^{m} \tilde{x}_{i,n} \right).$$

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We **center the dataset** by subtracting $\boldsymbol{\mu}$ from every row of $\widetilde{\boldsymbol{X}}$:

$$\boldsymbol{X} = \widetilde{\boldsymbol{X}} - \begin{pmatrix} \boldsymbol{\mu} \\ \boldsymbol{\mu} \\ \vdots \\ \boldsymbol{\mu} \end{pmatrix} \in \mathbb{R}^{m\times n},$$

so that each column of $\boldsymbol{X}$ has zero mean.

Sample Covariance Matrix

Given the centered data matrix $\boldsymbol{X} \in \mathbb{R}^{m \times n}$, the **sample covariance matrix** $\boldsymbol{C} \in \mathbb{R}^{n \times n}$ is:

$$\boldsymbol{C} = \frac{1}{m-1} \boldsymbol{X}^T \boldsymbol{X} = \begin{pmatrix} \operatorname{Cov}(A, A) & \operatorname{Cov}(A, B) & \cdots \\ \operatorname{Cov}(B, A) & \operatorname{Cov}(B, B) & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}.$$

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- **Diagonal entries** $C_{jj} = \mathrm{Var}(\text{feature } j)$ are the variances of the individual features.
- **Off-diagonal entries** $C_{jk} = \mathrm{Cov}(\text{feature } j, \text{feature } k)$ are the covariances between pairs of features.

PCA via Eigen-Decomposition and SVD

Since $\boldsymbol{C} = \frac{1}{m-1}\boldsymbol{X}^T\boldsymbol{X}$ is symmetric positive semi-definite, it admits an orthogonal spectral decomposition:

$$\boldsymbol{C} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^T, \qquad \boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n), \quad \lambda_1 \geq \cdots \geq \lambda_n \geq 0.$$

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- **Principal Axes (Directions):** Right singular vectors of $\boldsymbol{X}$ (columns of $\boldsymbol{V} = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$) are the eigenvectors of $\boldsymbol{C}$.

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- **Eigenvalues (Variances):** $\lambda_i = \dfrac{\sigma_i^2}{m-1}$ (SVD avoids forming $\boldsymbol{X}^T\boldsymbol{X}$!).

Multiplying $\boldsymbol{X} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ on the right by $\boldsymbol{V}$ gives the projected data matrix:

$$\boxed{\boldsymbol{X}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{\Sigma} = (\sigma_1 \boldsymbol{u}_1,\ \sigma_2 \boldsymbol{u}_2,\ \ldots,\ \sigma_n \boldsymbol{u}_n) \in \mathbb{R}^{m \times n}.}$$

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$\boldsymbol{X}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{\Sigma}$ is the **change of basis** of the rows of $\boldsymbol{X}$ into the orthonormal basis $\boldsymbol{V}$:

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Dimensionality reduction to $k < n$ features: Keep only the first k columns:

$$\boldsymbol{X}_k = \boldsymbol{X}\boldsymbol{V}_k = \boldsymbol{U}_k\boldsymbol{\Sigma}_k = (\sigma_1\boldsymbol{u}_1, \ldots, \sigma_k\boldsymbol{u}_k) \in \mathbb{R}^{m\times k}.$$

Total Variance and Explained Variance Ratio

Since the trace is invariant under orthogonal similarity transformations ($\mathrm{tr}(\boldsymbol{C}) = \mathrm{tr}(\boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^T) = \mathrm{tr}(\boldsymbol{\Lambda})$), the **total variance** T of the dataset is preserved:

$$T = \mathrm{tr}(\boldsymbol{C}) = \sum_{i=1}^{n} \mathrm{Var}(\text{feature } i) = \sum_{i=1}^{n} \lambda_i = \frac{1}{m-1} \sum_{i=1}^{n} \sigma_i^2.$$

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Explained Variance Ratio

The fraction of the total variance captured by the i-th principal component $\sigma_i \boldsymbol{u}_i$ (along axis $\boldsymbol{v}_i$) is:

$$\frac{\lambda_i}{T} = \frac{\lambda_i}{\sum_{j=1}^{n} \lambda_j} = \frac{\sigma_i^2}{\sum_{j=1}^{n} \sigma_j^2},$$

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Example 1: Nearly Uncorrelated 2D Features ($T = 100$)

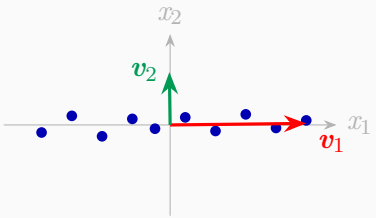
$$\boldsymbol{C}_1 = \begin{pmatrix} 95 & 1 \\ 1 & 5 \end{pmatrix}$$

Eigenvalues and principal axes:

$$\lambda_1 \approx 95, \qquad \lambda_2 \approx 5,$$

$$\boldsymbol{v}_1 \approx \begin{pmatrix} 0.999 \\ 0.011 \end{pmatrix}, \qquad \boldsymbol{v}_2 \approx \begin{pmatrix} -0.011 \\ 0.999 \end{pmatrix}.$$

The first principal component $\boldsymbol{v}_1$ captures $\frac{\lambda_1}{T} \approx \mathbf{95}\%$ of the total variance.



Example 2: Strongly Correlated 2D Features ($T = 100$)

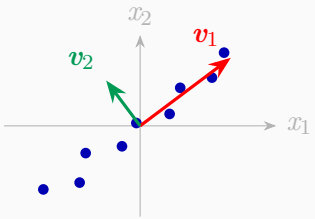
$$\boldsymbol{C}_2 = \begin{pmatrix} 50 & 40 \\ 40 & 50 \end{pmatrix}$$

Characteristic equation $(50 - \lambda)^2 - 40^2 = 0$:

$$\lambda_1 = 90, \qquad \lambda_2 = 10,$$

$$\boldsymbol{v}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \boldsymbol{v}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Projecting onto the $45°$ axis $\boldsymbol{v}_1$ captures $\frac{90}{100} = \mathbf{90}\%$ of the total variance in 1D!



Feature Standardization (z-Score)

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z-Score Standardization: Divide each centered feature j by its standard deviation $s_j = \sqrt{\mathrm{Var}(\text{feature } j)}$:

$$z_{ij} = \frac{\tilde{x}_{ij} - \mu_j}{s_j} \qquad \Longleftrightarrow \qquad \boxed{\boldsymbol{Z} = \boldsymbol{X}\boldsymbol{D}^{-1}}, \quad \boldsymbol{D} = \mathrm{diag}(s_1, \ldots, s_n).$$

Each column of $\boldsymbol{Z}$ has 0 mean and variance 1.

Covariance vs. Correlation Matrix in PCA

1. Covariance Matrix (Centered Data $\boldsymbol{X}$)

$$\boldsymbol{C} = \frac{1}{m-1}\boldsymbol{X}^T\boldsymbol{X}, \qquad T = \operatorname{tr}(\boldsymbol{C}) = \sum_{j=1}^{n} s_j^2.$$

Use when all features share the **same units and comparable scales**.

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Use when all features share the **same units and comparable scales**.

2. Correlation Matrix (z-Scored Data $\boldsymbol{Z} = \boldsymbol{X}\boldsymbol{D}^{-1}$)

$$\boldsymbol{R} = \frac{1}{m-1}\boldsymbol{Z}^T\boldsymbol{Z} = \boldsymbol{D}^{-1}\boldsymbol{C}\boldsymbol{D}^{-1}, \qquad R_{jj} = 1, \quad T = \operatorname{tr}(\boldsymbol{R}) = n.$$

Use when features have **different units or scales**.